Predicting Judicial Decisions: A Statistically Rigorous Approach and a New Ensemble Classifier

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***Abstract*—Natural language processing and machine learning are gaining wide popularity in supporting judicial decision-making. Research in this area is particularly active. However, a methodological issue in the use of AI methods can lead to poor statistical soundness in the results. We consider and improve the work of Aletras et. al. [1] for predicting the outcome of cases at the European Court of Human Rights. We replicate their experiments using a more statistically reliable methodology and analyzed the results using state-of-the-art Bayesian techniques for classifier comparison. We also improved classification accuracy using an ensemble-based approach. These techniques will widely improve the statistical soundness of machine learning applications in law by providing robust baselines for comparison.**

***Index Terms*—juridical decisions, natural language processing, machine learning, ensemble classifiers, classifiers comparison**

I. Introduction

In 1963 Lawlor predicted that machines would one day be able to predict judicial decisions [2]. A recent study considered this topic deeply [3]. AI systems for judicial decision-making can complement human judges allowing for greater efficiencies [4]. We focus on predicting the outcome of cases from the *European Court of Human Rights*. These cases are concerned with possible violations of the *European Convention on Human Rights* (ECHR). Applications are typically submitted by individuals against some state actor. Our objective is to improve the performance and robustness of text-mining and decision-making methods in order to provide a more reliable tool. The techniques we consider rely on a quantitative analysis of words and phrases used in court cases. We apply machine learning techniques to help a computer 'understand' the relationship between this information and the judicial outcome, to better support these decisions or even predict them.

Predicting judicial decisions is not a new topic of study. Early work focused on non-textual information, such as the gravity of the crime or on procedural irregularities [5]. Segal attempted to predict and explain the outcome of the cases of the American Supreme Court with mathematical regression [6]. A more recent study made use of information about the substantive similarity amongst cases to estimate judicial preferences that vary across substantive legal issues which is then used to predict judicial outcomes [7]. Many quantitative approaches to analyzing case law have been developed involving cases from many countries [8]–[10]. Branting classifies these studies as data-centric, as compared with logic-based approaches [11].

The ECHR has attracted much research in the use of data-driven methods for judicial outcome prediction [12]–[15]. For example, [16] used topic modelling techniques to automatically find latent topics in a set of judgements from the European Union Court of Justice Union and the European Court for Human Rights (ECtHR). As mentioned before, [1] proposed an approach to predict the outcome of cases of the ECtHR using natural language processing (NLP) followed by a classifier. That work inspired other researchers to work on the same data: for example, [17] increased the number of cases and articles involved and analysed different aspects of the same dataset, and [18] tested different classifiers.

However, our goal is not to develop a tool that can replace professional judges. Instead, we wish to stimulate research in this field to apply statistically sound methods and evaluate different approaches for this complex classification task. Without fair comparison we risk having the illusion of progress that has been observed in other fields [19], where new techniques claim to out-perform the state-of-the-art without real statistically significant improvement.

This paper is organized as follows. Section II discusses some methodological issues associated with earlier work in the field [1]. Section III presents a description of the dataset used in this work. Section IV replicates the results of [1] and analyzes them with modern classifier comparison methods. Section V shows the performance of our new classifier, which outperforms the approach reported in [1]. Finally, we conclude the paper and propose some future directions of study.

II. Limits of Earlier Work: A Case Study

The work of Aletras et al.[1] has been highly influential in the field of AI and law. However, the work has some methodological issues that have been propagated through the literature [8], [9], [17], [18], [20]. One of the aims of our work is to review these issues from a more rigorous methodological perspective.

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The two main technical issues in [1] are as follows. Firstly, the authors appear to only have applied 10-fold cross-validation once. The best-practice methodology is to average over m runs of k-fold cross-validation [21]–[23] and before each run the dataset should be shuffled. Typical values for m and k are 10 leading to 100 accuracy measures for each experiment. The shuffling step can strongly affect the results. For example, in our experimental results the average accuracy of the 10-fold cross-validation experiment can vary up to 9% between different shuffles.

Secondly, classifiers using the same algorithm but different preprocessing techniques can be considered as different and independent classifiers, and they can be evaluated as such. In this context, the comparison of classifiers based on average accuracy is not statistically sound. Modern statistical analysis offers us more advanced techniques based on the m runs of k-fold cross-validation [19], [22], [23]. In [1] the authors say: *"We observe that Circumstances is the best subsection to predict the decisions for cases in Articles 6 and 8, with a performance of .82 and .77 respectively."* While each of these statements is true, they must be considered carefully in order to draw statistically sound conclusions from the experimental data. In a similar study, [18] compares k-NN, logistic regression, bagging, random forests, and SVM, on the same preprocessed data. The final conclusions of that work are that an SVM outperforms the others due to its feature selection capability. More recent work increased the number of articles from the legislation as well as the number of cases per article [17]. Moreover, they added two new kinds of experiment. The first aims to predict future cases by dividing the cases used for training and testing on the basis of the year of the case. The second aims to forecast the outcome just using a judge's name. Even in that paper only a single run of 10-fold cross-validation was applied and accuracy, without standard deviation, was used as the only basis for comparison. It contains an impressive literature review, and the full methodology is described in a comprehensive manner, making it fully understandable even by non-technical readers.

Therefore, in each of these works the average accuracy based on a single run of cross-validation is used as a basis for comparison. This metric alone is not significant from a statistical point of view. It can strongly vary by simply reshuffling the dataset, as we show in Section IV.

III. Method

Our method comprises the following. First, we replicate the results reported in [1] and analyse them by using a variety of statistical techniques. Second, we introduce a new way to make judicial decisions based on documents with different subsections, motivated by work from ensemble classifiers. Third, we extend the previous technique to multiple classifiers. We hope that our methodology will become widely used in future research in this area. In addition, we believe that the ensemble technique we propose will be utilized in other real-world settings.

A. Classifier Comparison

The machine learning community, as well as other scientific areas, adopted the use of null hypothesis significance testing (NHST) in order to ensure the statistical validity of results [19], [21], [24]. On the other hand, NHST has proved to be inadequate for many reasons [25]. Bayesian inference has been used to improve the statistical soundness of hypothesis testing [26], [27]. A comprehensive analysis of the topic is presented in [22] which also introduces a Bayesian analysis of classifier performance based on *Bayesian parameter estimation* [28]. In this analysis a range of candidate values has to be set up, including the zero effect, and then Bayesian inference is used to compute the relative credibility of all candidate values, i.e. the posterior distribution. They use a useful concept, the *region of practical equivalence* (ROPE), introduced by [28]. In classification, it is sensible to regard two classifiers whose mean difference in accuracy is less than 1% as practically equivalent, this region represents that interval. This approach embraces the concept of statistical significance of the results in real applications.

We decided to utilize the *Bayesian correlated t-test* presented in [22] to give statistical soundness to the experimental part of this work. It is a non-parametric test that allows us to evaluate the possibility of two classifiers being *practically equivalent*. The Bayesian analysis proposed computes three posterior probabilities. Comparing Classifier A against Classifier B it computes:

- $P_{A>>B}$: Posterior probability of the mean difference of accuracies being practically in favour of Algorithm A;
- $P_{A==B}$ or P_{rope}: the posterior probability of the two classifiers being practically equivalentr;
- $P_{A<<B}$: posterior probability of the mean difference of accuracies being in favour of Algorithm B.

with $P_{A>>B} + P_{A==B} + P_{A<<B} = 1$. Such probabilities convey meaningful information even when they do not exceed the 95% threshold: this is a more informative outcome than that of a null hypothesis test [23]. This test takes account of correlations among the folds and the repetitions.

B. Description of the Datasets

We used the same datasets as those presented in [1]. These datasets are defined by a set of features extracted from cases from the European Court of Human Rights (ECtHR). The court has jurisdiction to rule on the possible violations of the European Convention of Human Rights (ECHR). The datasets relate to cases associated with violations of Article 3 (Prohibiting torture, inhuman and degrading treatment, 250 cases), Article 6 (Protecting the right to a fair trial, 80 cases) and Article 8 (Providing a right to respect one's private and family life, his home and his correspondence, 254 cases) of the Convention. To have an equivalent number of violation/non-violations cases the larger class of each datasets has been randomly reduced.

The judgment court cases have a distinctive structure. Cases are divided into a strict number of sections and subsections. The ones analysed by [1] are as follows:

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- *Procedure*: this contains the procedure followed by the court, from the lodging of the individual application until the judgment was handed down.
- *Circumstances*: this contains the factual background of the case and the procedure followed by the domestic courts. In this part are contained the applicant's story in its dealing with the respondent state's authorities.
- *Relevant Law*: this contains all legal provisions other than the articles of the Convention that can be relevant to deciding the case.
- *Law*: the vector standing for the law section considering the merits of the case through the use of legal argument.

The text features extracted from these cases are of two types:

- *N-gram features*: Using the Bag-of-Words (BOW) model the different subsections are represented as numeric vectors. The BOW approach is a popular semantic representation in natural language processing. It selects the most frequent N-grams (group of words) without taking into account grammar, syntax, or word order. For each dataset, they computed the top 200 most frequent N-grams where N $\in \{1, 2, 3, 4\}$. These features have been extracted for the four sections presented above, for the *facts* section (the union of *circumstances* and *relevant law*), and for the *full* document.
- *Topics*: N-grams that are semantically similar are clustered using spectral clustering [29]. These vectors represents the 30 most frequent topics in the court cases.

For a detailed description of the datasets and the NLP techniques applied we point the reader to the literature [17]. In [1] a combination of the two feature types has been proposed. Since the subsection with the highest performance of the N-grams feature was *circumstances*, they decided to combine it with the *topics* simply by concatenating the features. This type of feature is called "topics and circumstances".

C. Environment

We aimed to make all our results fully reproducible. Our R code is available online.[1] Every run of the algorithm requires a seed, that will generate all the subsequent pseudo-random numbers in the same order. All the classifiers are part of the `caret` package [30]. To compute the NBST we used the package `rNBST` [31]. As per the previous work we did not take into account computational times, so we do not specify the hardware configuration used. We tuned the parameters with a grid search.

IV. Analysis from Aletras et al.

We recreated the same experimental results as presented by [1]. We focused only on the classification part, without analyzing the process of mining the textual features from the cases. Before every 10-fold cross-validation we shuffled the dataset using one of the seeds provided. This operation has been repeated ten times. The 100 accuracy values are then averaged. We used the same classifier, SVM [32], but implemented in a different language. The only parameter to set is C; we used 100 as the previous work.

Table I contains the results of our experiments. Our results are comparable to the previous paper with some exceptions. For example, their best feature type is the *Topics and Circumstances*. They combined them for their high performances. However, in our test *Circumstances* is not performing better than the *Full* document, and its combination does not improve the average performances of *Topics*.

We believe that these differences are caused by a different implementation of the SVM, a different C value, and the 10-fold cross-validation approach.

With such small datasets doing multiple runs of cross-validation with reshuffling is very important. Table II shows the maximum difference in accuracy between the ten different runs. We can see that on average just by reshuffling we can obtain results with a 4% spread, and this value can grow if we add more shuffles. Therefore, it is possible to change the results significantly just by shuffling the dataset in the right way. The shuffle alone could have significantly impacted the conclusions in [1]. These differences can be statistically significant and averaging over 10 runs mitigates this effect.

The authors of [1] draw the following conclusion: *"Topics obtain the best performance in Article 3 and performance comparable to Circumstances in Articles 6 and 8."* Regarding the comparison with the *Circumstances* subsection, Table III presents the percentage of posterior probability contained in the three areas of the plots. According to [22] these are the percentages that one technique is better than the other or that they are practically equivalent. We can see that in Article 3 *Topics* outperforms *Circumstances*, however regarding the other two articles, no clear decision can be made. This is particularly due to the small size of the datasets. Therefore, we can confirm that the conclusion quoted above is partially correct, even if instead of *"comparable"* it might have been better to state that *"no clear decision can be made."*

The authors also state that: *"In Article 3, we obtain better predictive accuracy (.70) using the text extracted from the full case (Full) while the performance of Circumstances is almost comparable (.68)."* We can confirm that *Topics* does give the

TABLE I
AVERAGE CLASSIFICATION ACCURACY OF THE 10 RUNS OF 10-FOLD CROSS-VALIDATION. ALETRAS ET. AL. RESULTS BETWEEN BRACKETS.

Feature Type	Article 3	Article 6	Article 8	Average
Full	70.3 (70)	84.2 (82)	76.1 (72)	76.9 (75)
Procedure	65.6 (67)	76.6 (81)	65.0 (71)	69.1 (73)
Circumstances	71.3 (68)	82.9 (82)	75.9 (77)	76.7 (76)
Relevant Law	70.6 (68)	75.6 (78)	73.7 (72)	73.3 (73)
Facts	71.6 (70)	81.9 (80)	75.1 (68)	76.2 (73)
Law	48.3 (56)	67.9 (68)	61.1 (62)	59.1 (62)
Topics	**75.5** (78)	81.5 (81)	76.2 (76)	77.7 (78)
Topics and Cir	73.8 (75)	81.1 (84)	**77.6** (76)	77.5 (79)
Voting	72.9	**86.1**	77.4	**78.8**

[1] https://github.com/andvise/ECHR

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TABLE II
MAXIMUM ACCURACY DIFFERENCE BETWEEN SHUFFLES.

Feature Type	Article 3	Article 6	Article 8	Average
Full	2.4	2.5	3.2	2.7
Procedure	5.6	5.0	6.0	5.5
Circumstances	3.2	5.0	1.9	3.4
Relevant Law	2.4	3.7	5.5	3.9
Law	9.2	2.5	1.6	4.4
Topics	4.4	5.0	3.2	4.2
Facts	4.4	3.7	4.7	4.3
Topics and Cir	3.6	5.0	3.2	3.9
Average	4.4	4.1	3.7	4.0

TABLE III
PERCENTAGES OF POSTERIOR DISTRIBUTION OF *Circumstances* AGAINST *Topics*

	Circumstances	Equivalent	Topics
Article 3	3%	9%	88%
Article 6	54%	19%	27%
Article 8	29%	31%	40%

best performance in Article 3 with a probability of this being true of 90%, and the probability that this difference has a practical impact (part of the distribution on the right of the rope zone) of 83%. On the contrary, we cannot say that its performance is comparable to *Circumstances*. In Table III we can see that with 88% confidence, *Topics* is significantly better than *Circumstances* in Article 3. This is a more statistically sound way to draw conclusions from classifications results.

V. AN NEW ENSEMBLE CLASSIFIER

A. Voting classifier

The main idea behind the "Topics and Circumstances" dataset was to join two well-performing datasets to improve accuracy. All seven feature types (full, procedure, circumstances, relevant law, facts, law and topics) are different sets extracted from the same dataset. A set of SVMs, each trained in one feature type, can be considered seven different classifiers. The aim of this section is to combine them into an ensemble to achieve a better and more stable classifier [33]. Combining multiple weaker classifiers to build a better one is a common practice in machine learning [34].

The principle behind this approach is that a new classifier built on top of a set of independent decision makers can achieve higher accuracy compared to the individual ones. This is valid only if each classifier has an accuracy better than a baseline one, i.e. 50%. Out of the classifiers presented herein, only those built from the subsections can be considered as fully independent. We combine these using a simple majority voting scheme [35]. The outcome with most votes will be the winning one since their number is odd a tiebreaker is not needed. Table IV shows that the average performance of the *voting* classifier is better than the individual ones.

TABLE IV
PERCENTAGES OF POSTERIOR DISTRIBUTION OF *voting* AGAINST WELL-PERFORMING SECTIONS.

	Other	ROPE	Voting
Full Art 3	2.9	16.9	**80.3**
Full Art 6	7.6	25.5	**66.9**
Full Art 8	6.9	36.0	57.0
Circumstances Art 3	10.4	27.6	**62.0**
Circumstances Art 6	6.7	14.6	**78.8**
Circumstances Art 8	5.0	32.7	**62.4**
Topics Art 3	**72.8**	18.8	8.4
Topics Art 6	4.1	8.9	**87.1**
Topics Art 8	16.7	31.1	52.3
Top and cir Art 3	48.7	30.3	21.0
Top and cir Art 6	3.5	7.7	**88.8**
Top and cir Art 8	35.2	36.6	28.2

Table IV contains the posterior probabilities computed using the Bayesian Correlated t-test. We can see that the voting classifier can be considered an improvement that is significant over *Full* and *Circumstances*. In Article 8 the probability of them being practically equivalent is relevant. However, it is possible to exclude the fact that the original classifier was better (6.9 % and 5%).

On the contrary, *Topics* performs better than the others using Article 3. The difference with the *Voting* classifier is significant (72.8%). This can be justified by the fact that *Topics* contains most of the information useful to correctly classify the case and the individual sections perform poorly, especially *Law*, which is the only one having an accuracy lower than a baseline classifier (48.3% of average accuracy on Article 3). On the contrary *Voting* clearly outperforms all the other approaches using Article 6.

B. Performance of the Ensemble

After the interesting results of the first ensemble, we decided to extend this idea to further improve performance. Different algorithms exploit the data in different ways, with the result that a given algorithm may significantly outperform another on one problem instance. Classifiers, other than than SVM, can outperform it on average or in particular subsection. In this section, we want to repeat the work done in Section V-A with different classifiers, and then assign as a final decision the most common among those. It can be considered as an ensemble of ensembles. For simplicity, we will call it *ensemble* from now on. We selected two more classifiers: *gradient boosting machine (GBM)* and *K-nearest neighbour (KNN)*.

Table V shows the average accuracy for the three different voting classifiers based on the three algorithms. With all the classifiers the *Voting* subsection had the highest average accuracy. The *ensemble* has a slightly better average performance than the best of the single classifiers. However, this difference is not statistically significant. We computed all the posteriors of the *ensemble* classifier against SVM and for none of the

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TABLE V
AVERAGE ACCURACY OF THE *ensemble*.

Classifier	Article 3	Article 6	Article 8	Average
SVM	72.9	**86.1**	77.4	78.8
GBM	73.8	83.8	75.1	77.6
KNN	69.6	80.1	77.1	75.6
Ensemble	**73.9**	85.1	**78.0**	**79.0**

three articles is the probability higher than 60%. This makes it impossible to draw a clear conclusion, mostly due to the small size of the datasets.

VI. CONCLUSION AND FUTURE WORK

We considered a statistically rigorous method to compare classifiers in a law case-study. We hope that the field quickly adopts a more rigorous statistical evaluation technique, to avoid the illusion of progress in classification that has been observed in other fields [25].

We applied an ensemble approach to improve classification accuracy. We extended this method to different classifiers and different combination techniques. The results are promising and outperform the state-of-the-art. In our future work we aim to apply the techniques introduced herein on different datasets. A deeper analysis of the feature extraction techniques could be useful to generate different classifiers to add to an ensemble. A study of which classifiers can better perform in an ensemble and a deeper analysis of the search of hyperparameters would also be worthwhile.

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